EE 66 Fall 2026 — Signals, Dynamics, and Information — Homework 9

This homework is due Friday, October 30, 2026, at 23:59.

Submission Format

Your homework submission should consist of a single PDF file that contains all of your answers (any handwritten answers should be scanned). as well as your IPython notebook saved as a PDF.

On occasion, a problem set contains one or more problems designated as optional. Your solutions to such problems are not to be evaluated, and will not count toward your grade. Nevertheless, you are still responsible for learning the subject matter within their scope. In other words, we may test you on their scope.

1. A Change of Clock

Alok and Baris are watching a cart rolling along a straight track. Its position in meters (y_j) is measured at the times $t_j = j$ seconds ($j = 1,2,3,4,5$):

time t_j (s)	1	2	3	4	5
position y_j (m)	1	3	2	5	4

Alok and Baris measure the time with their own stopwatches, then fit a straight line to their data. Alok records the times above: $\mathbf{t} = (t_1,\ldots,t_5)^\top$ and $\mathbf{y} = (y_1,\ldots,y_5)^\top$. Let $\mathbf{1} = (1,1,1,1,1)^\top$.

Alok then fits the model $y \approx c_0 + c_1 t$ by least squares, minimizing $\left\|\mathbf{Ac}-\mathbf{y}\right\|^2$, where $\mathbf{A} = \begin{bmatrix}\mathbf{1} & \mathbf{t}\end{bmatrix}$ and $\mathbf{c} = (c_0,c_1)^\top$.

Baris started his stopwatch 3 seconds late (he reads $s = t-3$). Baris fits the model $y \approx d_0 + d_1 s$ to his measurements by minimizing $\left\|\mathbf{Bd}-\mathbf{y}\right\|^2$, where $\mathbf{B} = \begin{bmatrix}\mathbf{1} & \mathbf{s}\end{bmatrix}$, $\mathbf{s} = \mathbf{t} - 3\cdot\mathbf{1}$, and $\mathbf{d} = (d_0,d_1)^\top$.

(a) Determine Baris's least-squares estimates d_0 and d_1. You should <u>not</u> need to invert any non-diagonal matrix.

(b) Explain why Alok and Baris obtain the same fitted vector $\widehat{\mathbf{y}}$. Use this to show that

$$c_1 = d_1 \qquad \text{and} \qquad c_0 = d_0 - 3d_1,$$

and hence determine Alok's estimates c_0 and c_1.

(c) Let $\mathbf{e} = \mathbf{y} - \widehat{\mathbf{y}}$ denote the residual.

i. Use the normal equations to explain why $\mathbf{e} \perp \widehat{\mathbf{y}}$.

ii. Determine the minimum squared error $\|\mathbf{e}\|^2$. You should <u>not</u> need to compute the individual residuals.

(d) Now suppose the measurement times $t_1,\ldots,t_m$ are arbitrary, but not all equal, with mean $\bar{t} = \frac{1}{m}\sum_{j=1}^m t_j$; let $\bar{y} = \frac{1}{m}\sum_{j=1}^m y_j$. Show that the least-squares line $y \approx c_0 + c_1 t$ passes through the point $(\bar{t},\bar{y})$.

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2. SVD Computation

(a) Consider the matrix

$$\mathbf{A} = \begin{bmatrix} -1 & 1 & 5 \\ 3 & 1 & -1 \\ 2 & -1 & 4 \end{bmatrix}.$$

Observe that the columns of matrix $\mathbf{A}$ are mutually orthogonal with norms $\sqrt{14}, \sqrt{3}, \sqrt{42}$.

Verify numerically that columns $\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$ **and** $\begin{bmatrix} 5 \\ -1 \\ 4 \end{bmatrix}$ **are orthogonal to each other.**

(b) **Write $\mathbf{A} = \mathbf{BD}$, where B is an orthonormal matrix and D is a diagonal matrix.** What is **B**? What is **D**?

(c) **Write out a valid singular value decomposition of $\mathbf{A} = \mathbf{U\Sigma V}^\top$ using the result from the previous part.** Note that the singular values in $\mathbf{\Sigma}$ should be ordered from largest to smallest.
(HINT: There is no need to compute any eigenvalues.)

(d) Given a new matrix

$$\mathbf{A} = \frac{1}{\sqrt{50}} \begin{bmatrix} 3 \\ 4 \end{bmatrix} \begin{bmatrix} 1 & -1 \end{bmatrix} + \frac{3}{\sqrt{50}} \begin{bmatrix} -4 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix}, \tag{1}$$

write out a singular value decomposition of matrix A in the form $\mathbf{U\Sigma V}^\top$. Remember that the singular values in $\mathbf{\Sigma}$ should be ordered from the largest to smallest.
(HINT: You don't have to compute any eigenvalues for this. Some useful observations are that

$$\begin{bmatrix} 3 & 4 \end{bmatrix} \begin{bmatrix} -4 \\ 3 \end{bmatrix} = 0, \quad \begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 0, \quad \left\| \begin{bmatrix} 3 \\ 4 \end{bmatrix} \right\| = \left\| \begin{bmatrix} -4 \\ 3 \end{bmatrix} \right\| = 5, \quad \left\| \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\| = \left\| \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\| = \sqrt{2}.$$

)

(e) Let us define a new matrix

$$\mathbf{A} = \begin{bmatrix} -1 & 4 \\ 1 & 4 \end{bmatrix}.$$

Find the SVD of A by following the standard algorithm introduced in lecture (i.e. by computing the eigendecomposition of $\mathbf{A}^\top\mathbf{A}$). Also find the eigenvectors and eigenvalues of A. You may use a calculator to solve the roots of the characteristic polynomial *for the eigenvalues of* **A**, but not for the SVD. Is there a relationship between the eigenvalues or eigenvectors of **A** with the SVD of **A**?

3. Rank 1 Decomposition and Image Compression With SVD

In this problem, we will introduce an important application of the singular value decomposition (SVD) called low-rank approximation in the context of image compression. Given a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ of rank $r \leq \min\{m,n\}$, we know that we can write $\mathbf{A}$ using the outer-product form of the SVD:

$$\mathbf{A} = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top. \tag{2}$$

If our matrix is high-rank, i.e., $r \approx \min\{m,n\}$, then almost all the σ_i will be nonzero and non-negligible. However, if the data has some linear, low-rank essential structure, as is usually the case with real data such as images, most of our singular values will be very small (but usually nonzero due to noise or disturbances). If, say, the data has rank ℓ, then the first ℓ singular values are large, and the remaining $r-\ell$ are small:

$$\mathbf{A} = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top \tag{3}$$

$$= \sum_{i=1}^{\ell} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top + \sum_{i=\ell+1}^{r} \underbrace{\sigma_i}_{\approx 0} \mathbf{u}_i \mathbf{v}_i^\top \tag{4}$$

$$\approx \sum_{i=1}^{\ell} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top. \tag{5}$$

This motivates approximating the data as

$$\mathbf{A}_\ell := \sum_{i=1}^{\ell} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top \tag{6}$$

and using this compressed data for further analysis. As images are readily represented as matrices, in this problem, we will see how to make use of this low rank approximation technique to compress images without severely deteriorating the image quality.

Throughout this problem we will work in the Jupyter Notebook accompanying this homework. Part 1 of the notebook relates to finding rank 1 decompositions of images, while Parts 2 and 3 focus on image compression. Make sure to read through all the sections of the notebook to understand how we will use what we know about the SVD in order to compress images! You should write all your answers in the written submission and you don't need to submit the parts of the notebook related to this problem.

We will start by decomposing a few images into linear combinations of rank 1 matrices. Remember that outer product of two vectors $\mathbf{s}\mathbf{g}^\top$ gives a rank 1 matrix. It has rank 1 because the column span is one-dimensional — multiples of $\mathbf{s}$ only — and the row span is also one dimensional — multiples of $\mathbf{g}^\top$ only. More generally, this exercise should provide intuition into how the outer product form of the SVD decomposes the underlying matrix into a sum of constituent rank 1 matrices with different weightings given by the singular values.

(a) Consider a standard 4×4 checkerboard shown in Figure 1. Assume that black colors represent -1 and that white colors represent 1.

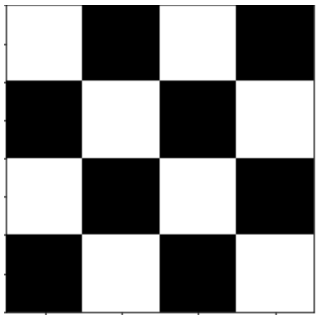


Figure 1: 4×4 checkerboard.

The checkerboard is given by the following 4×4 matrix $\mathbf{L}$:

$$\mathbf{L} = \begin{bmatrix} 1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 1 \\ 1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 1 \end{bmatrix} \tag{7}$$

Express L as a linear combination of outer products. There are many solutions, but providing one is enough. It is not necessary for the vectors to be normalized. You may use the helper functions in the accompanying Jupyter Notebook to check your answer.

(HINT: In order to determine how many rank 1 matrices you need to combine to represent the matrix, find the rank of the matrix you are trying to represent.)

(b) We next consider the same checkerboard given by the following 4×4 matrix $\mathbf{C}$ where black colors now represent 0 and that white colors represent 1.

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \tag{8}$$

Express C as a linear combination of outer products. There are many solutions, but providing one is enough. It is not necessary for the vectors to be normalized. You may use the helper functions in the accompanying Jupyter Notebook to check your answer.

(HINT: Note that $\mathbf{C}$ *is a rank 2 matrix and therefore you need 2 outer products to represent it. To find these individual outer products consider the columns of* $\mathbf{C}$*.)*

(c) Now consider the Swiss flag shown in Figure 2. Assume that red colors represent 0 and that white colors represent 1.

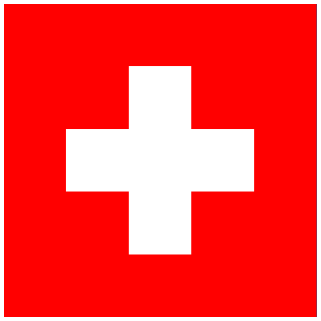


Figure 2: Swiss flag.

Assume that the Swiss flag is given by the following 5×5 matrix $\mathbf{S}$:

$$S = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{9}$$

Express S as a linear combination of outer products. There are many solutions, but providing one is enough. It is not necessary for the vectors to be normalized. You may use the helper functions in the accompanying Jupyter Notebook to check your answer.

(d) To represent the full $\mathbf{A} \in \mathbb{R}^{m \times n}$, we need $m \cdot n$ numbers. If we instead take the rank-k approximation $\mathbf{A}_k := \sum_{i=1}^{k} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$, **in terms of** $k, m,$ **and** n**, how many real numbers do we require to describe** $\mathbf{A}_k$**?**

(e) Here we show two images of a kitten: An uncompressed $\mathbf{A}$ with dimensions 900×1200 (right), and a compressed rank 200 approximation of $\mathbf{A}$ (left). **Can you easily spot any difference between the two images even though we are using only about 40% of the data?**

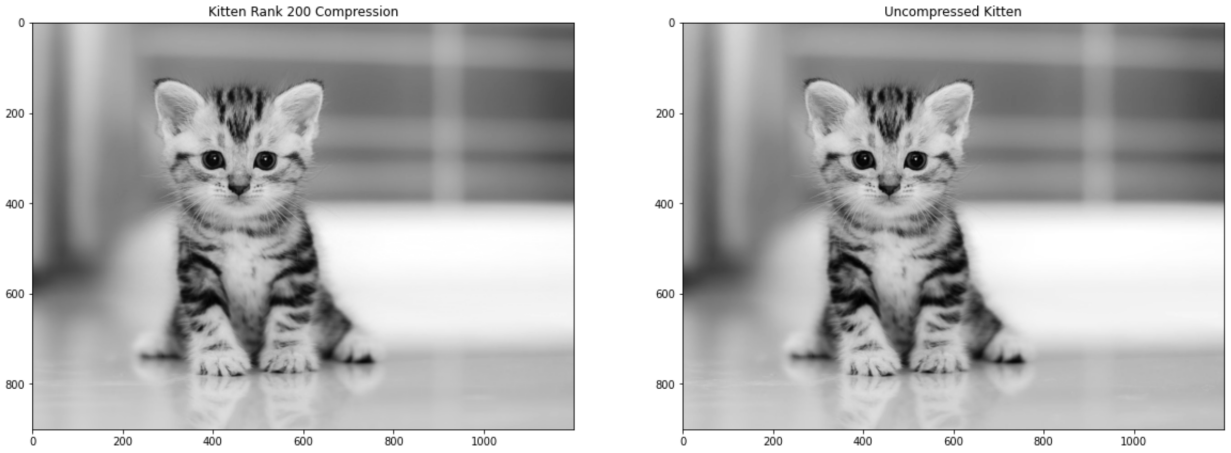


Figure 3: Compressed vs uncompressed image of a kitten.

(f) The accompanying Jupyter Notebook contains a slider that allows you to adjust the rank k of the approximation. **What is the lowest acceptable value of** k **that you can go without losing too much image quality? What percentage of the data is needed?**

4. SVD Proofs

Given the SVD of a matrix $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$, we can write $\mathbf{V} = \begin{bmatrix}\mathbf{V}_r & \mathbf{V}_{n-r}\end{bmatrix}$, where $\mathbf{V}_r$ are the first r columns of $\mathbf{V}$ and $\mathbf{V}_{n-r}$ are the last $n-r$ columns of $\mathbf{V}$. Similarly, we have $\mathbf{U} = \begin{bmatrix}\mathbf{U}_r & \mathbf{U}_{m-r}\end{bmatrix}$, where $\mathbf{U}_r$ are the first r columns of $\mathbf{U}$ and $\mathbf{U}_{m-r}$ are the last $m-r$ columns of $\mathbf{U}$.

(a) Show, for a general matrix $\mathbf{A} \in \mathbb{R}^{m\times n}$ with $\text{rank}(\mathbf{A}) = r$ and $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$, that $\text{Null}(\mathbf{A}) = \text{Col}(\mathbf{V}_{n-r})$. *HINT: How do we show that two spaces are equal? Consider using the outer product form of SVD. It may also be helpful to recall the rank-nullity theorem:* $\dim(\text{Col}(\mathbf{A})) + \dim(\text{Null}(\mathbf{A})) = n$.

(b) Show, for a general matrix $\mathbf{A} \in \mathbb{R}^{m\times n}$ with $\text{rank}(\mathbf{A}) = r$ and $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$, that $\text{Col}(\mathbf{A}) = \text{Col}(\mathbf{U}_r)$.

(c) Show, for a general matrix $\mathbf{A} \in \mathbb{R}^{m\times n}$ with $\text{rank}(\mathbf{A}) = r$ and $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$, that $\text{Null}(\mathbf{A}^\top) = \text{Col}(\mathbf{U}_{m-r})$ and $\text{Col}(\mathbf{A}^\top) = \text{Col}(\mathbf{V}_r)$. Use these facts and the previous subparts to show that $\dim(\text{Col}(\mathbf{A})) + \dim(\text{Null}(\mathbf{A}^\top)) = m$, and that $\text{Col}(\mathbf{A})$ and $\text{Null}(\mathbf{A}^\top)$ are orthogonal.

5. Conceptual PCA

(a) Consider a data matrix $\mathbf{A} \in \mathbb{R}^{d \times n}$, where n is the number of data points and d is the dimensionality of each data point. The data points, $\mathbf{a}_i$, are the columns of $\mathbf{A}$. One way to formulate PCA is to find an ℓ-dimensional subspace that minimizes the total squared reconstruction error:

$$\underset{\mathbf{W} \in \mathbb{R}^{d \times \ell}}{\operatorname{argmin}} \sum_{i=1}^{n} \left\| \mathbf{a}_i - \mathbf{W}\mathbf{W}^\top \mathbf{a}_i \right\|^2 \tag{10}$$

where $\mathbf{W}^\top \mathbf{W} = \mathbf{I}_\ell$. The columns of $\mathbf{W}$ are ℓ orthonormal directions spanning the chosen subspace. Multiplying by $\mathbf{W}^\top$ gives each data point's coordinates along these directions, and multiplying again by $\mathbf{W}$ maps those coordinates back to the original space. Thus $\mathbf{W}\mathbf{W}^\top \mathbf{A}$ is the reconstruction of our data we get using only the information captured by our subspace.

Since $\mathbf{W}$ has rank ℓ, the reconstruction retains only ℓ dimensions. For $\ell = 1$ (i.e., to find the first principal component), this can be rewritten as

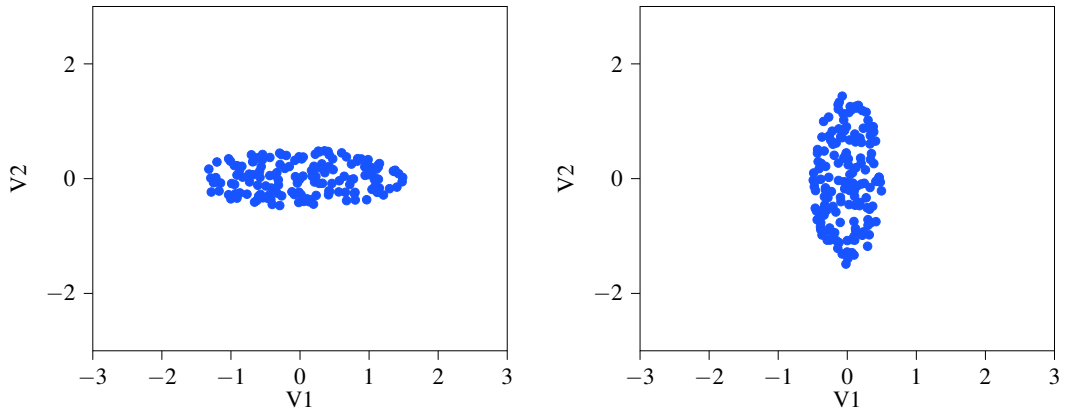
$$\underset{\mathbf{u} \in \mathbb{R}^d}{\operatorname{argmin}} \sum_{i=1}^{n} \|\mathbf{a}_i - \langle \mathbf{a}_i, \mathbf{u} \rangle \mathbf{u}\|^2 \tag{11}$$

where $\|\mathbf{u}\| = 1$. **Show that finding the top principal component is equivalent to maximizing**

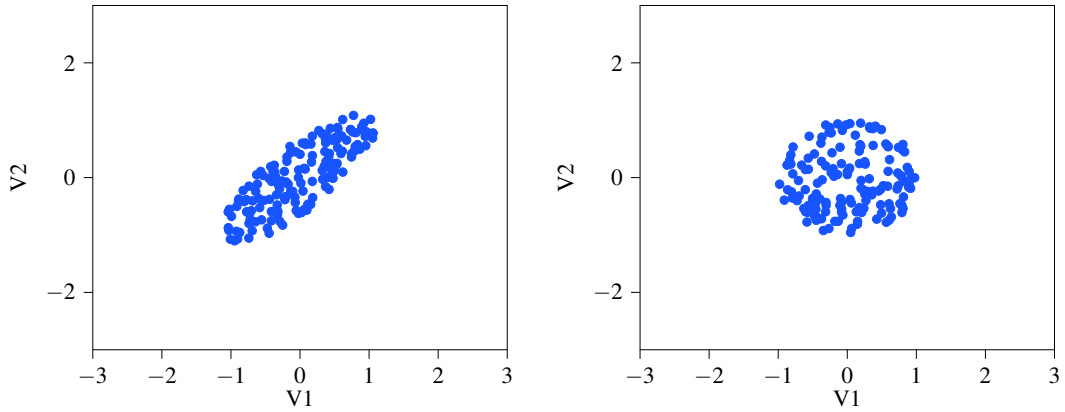
$$\sum_{i=1}^{n} \langle \mathbf{a}_i, \mathbf{u} \rangle^2 \tag{12}$$

In each plot below, the data is projected onto two unit vectors. The x coordinate is the projection onto the first vector (written as "V1" or $\mathbf{v}_1$), and the y coordinate is the projection onto the second vector (written as "V2" or $\mathbf{v}_2$). We say that a plot is "valid" if the first vector would be the first principal component, and if the second vector would correspond to the second principal component. For each subpart, explain your answer.

(b) **Which of these two plots is valid?**



(c) **Which of these two plots is valid?**



(d) Part (a) showed that finding the first principal component means maximizing $\sum_{i=1}^{n} \langle \mathbf{a}_i, \mathbf{u} \rangle^2$ over unit vectors $\mathbf{u}$. We will now connect this to the SVD of the data matrix.

Let $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ be the SVD of the data matrix, with singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$ and left singular vectors $\mathbf{u}_1, \ldots, \mathbf{u}_r$. **Show that evaluating the objective above at $\mathbf{u} = \mathbf{u}_k$ gives**

$$\sum_{i=1}^{n} \langle \mathbf{a}_i, \mathbf{u}_k \rangle^2 = \sigma_k^2. \tag{13}$$

Then explain what this says about which vector is the first principal component.

Hint: The sum is over the columns $\mathbf{a}_i$ of $\mathbf{A}$, so it collects the entries of the row vector $\mathbf{u}_k^\top \mathbf{A}$. What is $\mathbf{u}_k^\top \mathbf{A}$ in terms of the SVD?

6. Movie Ratings and PCA

For this problem, imagine that we can think of movie ratings as a structured set of data. For every person i and movie j, we have that person's rating $\mathbf{R}_{ij}$ (thought of as a real number).

Suppose that there are m movies and n people. Let's think about arranging this data into a big $n \times m$ matrix $\mathbf{R}$ with people corresponding to rows and movies corresponding to columns. So the entry in the i-th row and j-th column should be $\mathbf{R}_{ij}$ above. Note that this is organized differently from how it was in lecture. Each row corresponds to a unique person and each column to a unique movie.

$$\mathbf{R} = \begin{bmatrix} \mathbf{R}_{11} & \mathbf{R}_{12} & \cdots & \mathbf{R}_{1m} \\ \mathbf{R}_{21} & \mathbf{R}_{22} & \cdots & \mathbf{R}_{2m} \\ \vdots & \vdots & \cdots & \vdots \\ \mathbf{R}_{n1} & \mathbf{R}_{n2} & \cdots & \mathbf{R}_{nm} \end{bmatrix} \tag{14}$$

(a) Suppose we believe that there is actually an underlying pattern to this rating data and that a separate study has revealed that every movie is characterized by a set of characteristics: say action and comedy. This means that every movie j has a pair of numbers a_j (for action) and c_j (for comedy) that define it. At the same time, every person i has a pair of sensitivities f_i and g_i that define their preferences for action vs. comedy in movies respectively. A person i will rate the movie j as $\mathbf{R}_{ij} = f_i a_j + g_i c_j$.

If we arrange the sensitivities into a pair of n-dimensional vectors $\mathbf{f}, \mathbf{g}$ for our group of n people, and the movie characteristics into a pair of m-dimensional vectors $\mathbf{a}, \mathbf{c}$ for our group of m movies, **use outer products to express the rating matrix R in terms of these vectors $\mathbf{f}, \mathbf{g}, \mathbf{a}, \mathbf{c}$.**

(b) Now suppose that we want to discover the underlying nature of movies from the data $\mathbf{R}$ itself.

Suppose for this part that you have four observed rating data vectors (corresponding to four different movies being rated by six individuals).

All of the movie data vectors just happened to be multiples of the following 6-dimensional vector

$$\mathbf{w} = \begin{bmatrix} 2 \\ -2 \\ 3 \\ -4 \\ 4 \\ 0 \end{bmatrix}.$$

(For your convenience, note that $\|\mathbf{w}\| = 7$.)

You arrange the movie data vectors as the columns of a matrix $\mathbf{R}$ given by:

$$\mathbf{R} = \begin{bmatrix} | & | & | & | \\ -\mathbf{w} & -2\mathbf{w} & 2\mathbf{w} & 4\mathbf{w} \\ | & | & | & | \end{bmatrix} \tag{15}$$

You want to perform PCA (for movies) using the SVD of the matrix $\mathbf{R}$ to better understand the pattern in your data.

The first "principal component vector" is a *unit vector* that tells which direction we would want to project the columns of $\mathbf{R}$ onto to get the best rank-1 approximation for $\mathbf{R}$.

Find this first principal component vector of the columns of R to explain the nature of your movie data points.

(HINT: You don't need to actually compute any SVDs in this simple case. Also, be sure to think about what size vector you want as the answer. Don't forget that you want a unit vector!)

(c) Suppose that we now collected two more data points (corresponding to two more movies being rated by the same set of six people, i.e. two additional columns for our matrix) that are multiples of a different vector $\mathbf{p}$ where:

$$\mathbf{p} = \begin{bmatrix} 6 \\ 3 \\ -2 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

(For your convenience, note that $\|\mathbf{p}\| = 7$ and that $\mathbf{p}^\top \mathbf{w} = 0$.)

We augment our ratings data matrix with these two new data points to get:

$$\mathbf{R} = \begin{bmatrix} | & | & | & | & | & | \\ -\mathbf{w} & -2\mathbf{w} & 2\mathbf{w} & 4\mathbf{w} & -3\mathbf{p} & 3\mathbf{p} \\ | & | & | & | & | & | \end{bmatrix} \tag{16}$$

Find the first two principal components corresponding to the nonzero singular values of $\mathbf{R}$. This is what we would use to best project the movie data points onto a two-dimensional subspace.

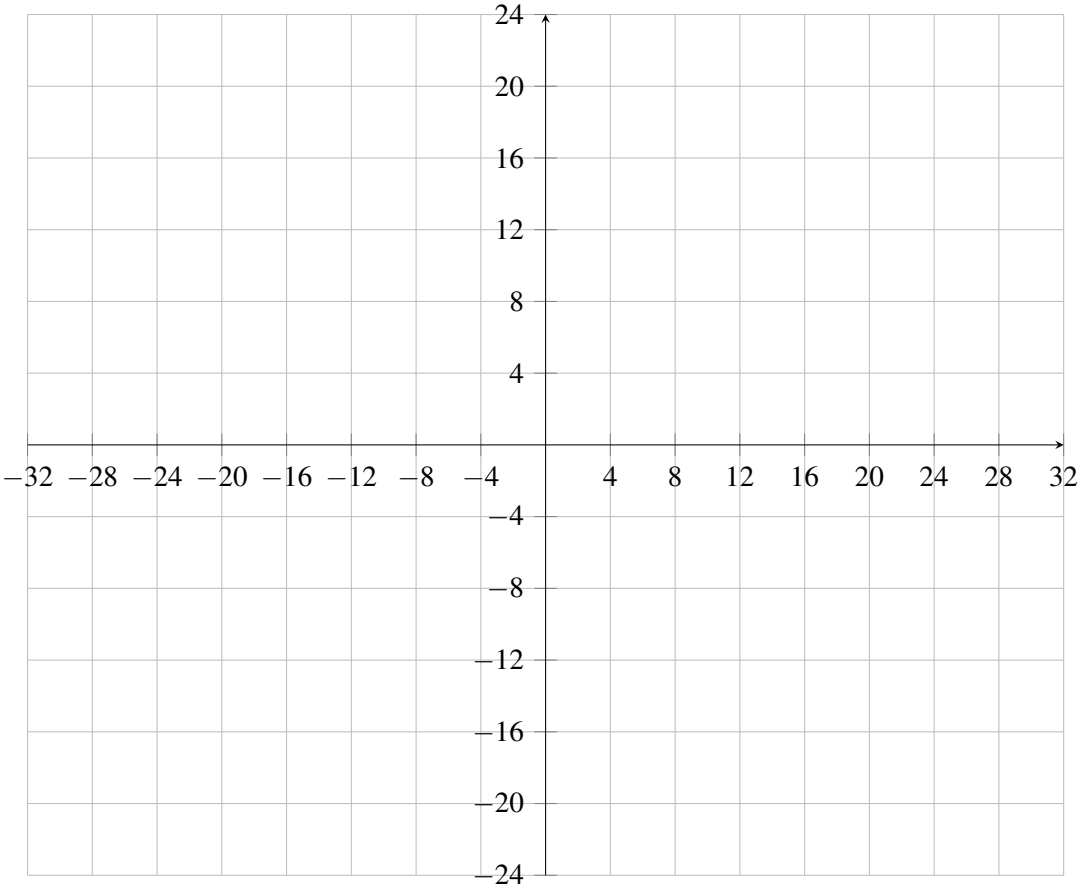
What is the first principal component vector? What is the second principal component vector? Justify your answer. *(HINT: Think about the inner product of* $\mathbf{w}$ *and* $\mathbf{p}$ *and what that implies for being able to appropriately decompose* $\mathbf{R}$*. Again, very little computation is required here.)*

(d) In the previous part, you had

$$\mathbf{R} = \begin{bmatrix} | & | & | & | & | & | \\ -\mathbf{w} & -2\mathbf{w} & 2\mathbf{w} & 4\mathbf{w} & -3\mathbf{p} & 3\mathbf{p} \\ | & | & | & | & | & | \end{bmatrix}$$

with $\|\mathbf{w}\| = 7$ and $\|\mathbf{p}\| = 7$, satisfying $\mathbf{p}^\top \mathbf{w} = 0$.

Plot the movie data points $\mathbf{r}_i$ (for all i, where $\mathbf{r}_i$ denotes the i-th column of R) projected onto the first and second principal component vectors of the columns of R. The horizontal axis should reflect the coordinate along the first principal component, and the vertical axis should reflect the coordinate along the second principal component. **Label each point, and the axes. Remember that principal component vectors are normalized**.


24
20
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−4
−8
−12
−16
−20
−24

7. Homework Process and Study Group

Whom did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.